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The man who stops advertising to save money is the man who saves the clock to stop time.

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-Henry Ford

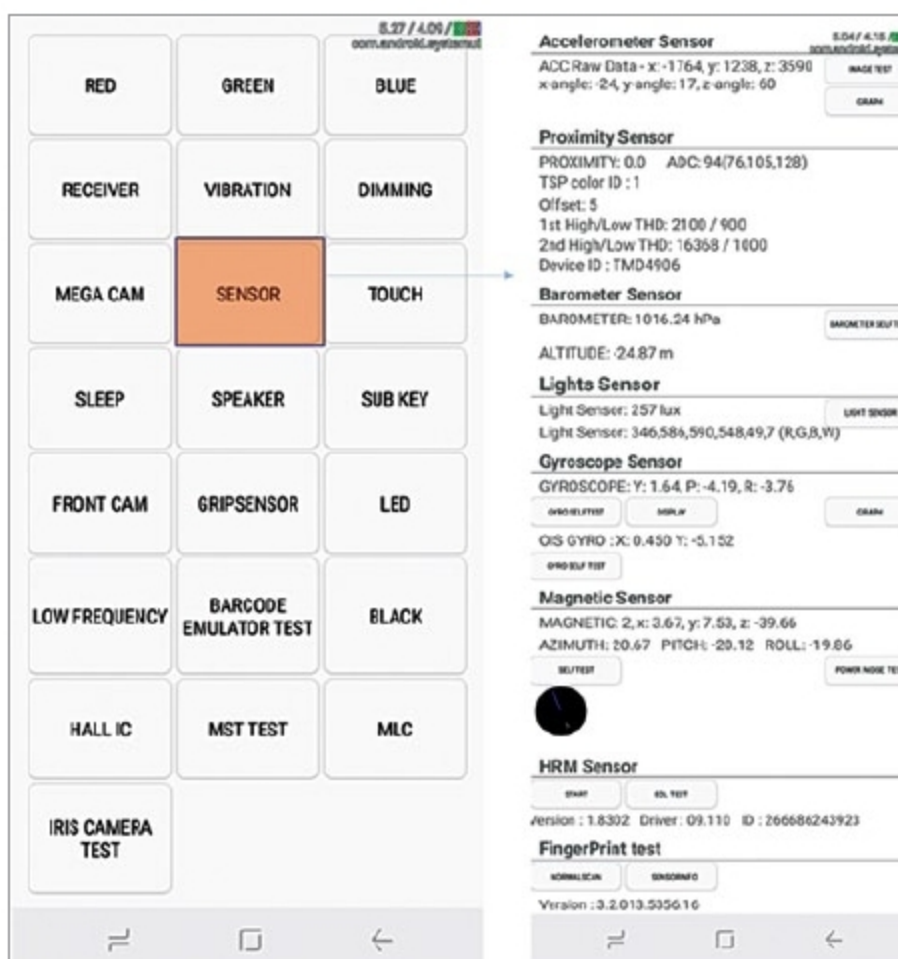


Fig. 2: Sensors on Samsung S8 device

other areas such as imaging, virtual reality (VR), augmented reality (AR), indoor positioning and password-free personal authentication.

However, this very sensitivity and power of these open the door for some heinous attacks to be carried out with much ease that were hitherto not achievable.

In the rest of this article, I survey the landscape of such attacks, categorise these and highlight the flagship examples in each category. The goal of this venture is to not just highlight these threats to the security community at large, but to also pique the interest of sensor designers and manufacturers who will play a crucial role in

certain frequencies, for example). This dirties up the very data that the sensor is picking up.

Attacks with malicious intent involve nefariously eavesdropping on sensor data and inferring things that the user might be typing or talking without having to access the traditional well-guarded sources of such information, like the microphone.

Now, let's dive deeper.

Spoofing. Spoofing attacks in the context of smartphone sensors are carried out in two ways: spoofing by injection and spoofing by transduction.

Spoofing by injection entails an attacker injecting spurious sensor data

overcoming these threats.

Most security attacks on sensors broadly fall into two categories: spoofing attacks and attacks with intent of malicious use. Spoof attacks actually target modification of sensor data being used by a legitimate application, either by cunningly injecting fake or synthetic data somewhere in the pipeline, or introducing adversarial vibrations around the phone (by playing sounds of

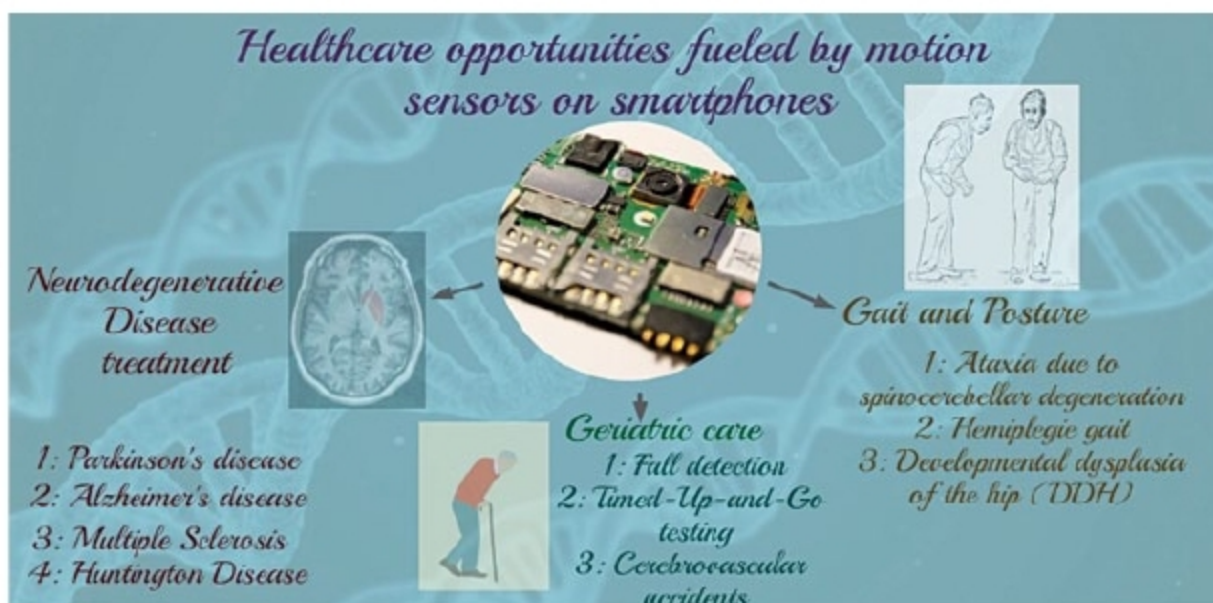


Fig. 3: Specific domains of personal healthcare being influenced by motion sensors